

Project Documentation: Mini-SIMS Portal

A Consolidated Student Information Management System

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1. Executive Summary

Mini-SIMS is a robust, full-stack Student Information Management System designed to consolidate daily school operations into a single, cohesive digital environment. Originally conceptualized as a static interface, the architecture was elevated to an enterprise-grade React application backed by a live PostgreSQL database. The platform unifies attendance tracking, fee management, academic scheduling, and administrative oversight, delivering a seamless User Experience (UX) across desktop and mobile devices.

2. Problem Statement & Solution

Traditional educational institutions often suffer from fragmented operational data. Attendance is tracked on paper, fees are managed in separate accounting software, and schedules are distributed manually.

Pain Points Addressed:

- **Data Fragmentation:** Mini-SIMS unifies attendance, fee tracking, and academic schedules into a single dashboard.
- **Manual Attendance & Proxy-Spoofing:** Solved via a secure, live IoT QR scanner module that validates dynamic student identifiers in real-time.
- **Lack of Administrative Oversight:** Solved by introducing a dynamic Rule-Based Risk Engine that automatically flags students at risk due to low attendance or unpaid fees.
- **Accessibility Barriers:** Solved through a fully mobile-responsive interface utilizing Tailwind CSS, allowing lecturers to input grades and take attendance directly from their mobile devices.

3. Technology Stack & Architecture

The system utilizes a modern, decoupled architecture, separating the client-side interface from the database and authentication layers.

- **Frontend Framework:** React.js
- **State Management:** Redux Toolkit (`authSlice`, `studentSlice`) with `localStorage` persistence.
- **Routing:** React Router DOM (with protected RBAC interceptors).
- **Styling & UX:** Tailwind CSS, Lucide React (Iconography).
- **Backend & Database:** Supabase (PostgreSQL).
- **Hardware Integration:** `html5-qrcode` and `qrcode.react` for IoT camera scanning.
- **Hosting:** Vercel (CI/CD integration with GitHub).

4. Role-Based Access Control (RBAC)

Security is enforced at the routing layer. An authentication interceptor evaluates the session state and forcefully routes users strictly to their authorized workspaces, preventing privilege escalation.

1. **Administrator:** Full oversight of student directories, fee ledgers, and institutional analytics.
2. **Lecturer:** Access to course-specific grade entry, manual registries, and live QR scanning sessions.
3. **Student:** Read-only access to personal academic transcripts, financial status, timetables, and generation of dynamic QR identity tokens.

5. Core Modules & Functionality

5.1. The Administrator Workspace

- **Rule-Based Risk Engine:** A custom algorithm that evaluates live database metrics. It assigns weighted penalty points for low attendance (e.g., < 75%) and pending financial statuses, instantly sorting students into "Critical," "Warning," or "Good" tiers for immediate administrative action.
- **Financial Orchestration:** Centralized tracking of total pending fees versus cleared accounts.
- **Student Directory:** A searchable, filterable database of all enrolled students featuring empty-state fallbacks and pagination-ready UI.

5.2. The Lecturer Workspace (Attendance & Grading)

- **IoT QR Scanner Module:** Transforms the lecturer's device into a biometric scanner. It reads dynamic QR tokens generated by students, communicating securely with a Supabase Remote Procedure Call (RPC) to validate course enrollment before marking attendance.
- **Manual Registry (Optimistic UI):** For students without devices, lecturers can manually toggle attendance. The interface employs Optimistic UI patterns, updating the screen instantly while the database synchronizes asynchronously in the background.
- **Grade Entry:** A secure portal for continuous assessment (CA) and exam score inputs, featuring automated letter-grade calculation and input boundary validation (CA max 30, Exam max 70).

5.3. The Student Workspace

- **Dynamic Identity Token:** Generates a high-correction QR code containing the student's unique registration payload, utilized for the reverse-authentication attendance system.
- **Academic Transcript:** A consolidated, printable view of course credits, letter grades, and a calculated CGPA.
- **Interactive Timetable:** Displays "Next Classes" dynamically, complete with visual conflict indicators.

6. Logic, Data Flow & Error Handling

Mini-SIMS moves away from standard mock JSON servers to demonstrate advanced data orchestration:

- **Redux Slices:** Global state is managed via Redux, ensuring that financial data, attendance records, and user sessions remain synchronized across diverse components without prop-drilling.
- **Asynchronous Operations:** Data fetching and database mutations are handled via Redux `createAsyncThunk`, communicating directly with the Supabase client.
- **Error Handling:** The application features robust API failure catchers. Invalid logins, database timeouts, or spoofed QR codes trigger immediate, user-friendly error banners and toast feedback mechanisms.
- **Accessibility (a11y):** Form inputs enforce HTML5 validation, and the DOM structure utilizes semantic HTML and ARIA-compliant patterns to support keyboard navigation.

7. Database Schema (PostgreSQL)

The backend enforces data integrity through a highly relational PostgreSQL structure:

- **students:** Stores core profiles, global attendance rates, and financial statuses.
- **courses:** Stores academic course codes, titles, and credit units.

- **enrollment**: A junction table enforcing unique constraints so a student cannot enroll in the same course twice.
- **live_attendance**: The immutable ledger for the IoT scanner. It utilizes `UNIQUE(student_id, course_id, date)` constraints to prevent duplicate attendance logs on the same day.

8. Conclusion

Mini-SIMS successfully delivers a cohesive, multi-module dashboard that mirrors real-world SaaS operations. By integrating hardware scanning capabilities, strict RBAC security, live PostgreSQL data synchronization, and an intelligent risk-scoring algorithm, the platform exceeds standard administrative portal requirements and stands as a robust, production-ready educational tool.